John Rugemalila

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Education

The University of Chicago Expected, May 2027
B.A. in Computer Science | GPA: 3.77 | Dean's List | Odyssey Scholar *Chicago, IL*

- **Relevant coursework:** Data Structures & Algorithms, Object-Oriented Programming, Distributed Systems, Machine Learning & AI, Software Development, Database Systems, Data Science, Discrete Mathematics

Skills

Languages: Java, Python, Go, Rust, TypeScript, JavaScript, C, SQL, GraphQL
Frameworks & Libraries: gRPC, FastAPI, Spring Boot, React, Next.js, PyTorch, scikit-learn, NumPy
Tools & Platforms: AWS, Snowflake, Kafka, PostgreSQL, MongoDB, ChromaDB, Firebase, Linux, Git, Docker

Experience

HubSpot June 2026 – August 2026
Software Engineer Intern *Boston, MA*

- Built a distributed async pipeline connecting data ingestion to privacy review, using task queues, idempotent Java workers, and Snowflake processing to eliminate privacy review gaps and guarantee 100% coverage
- Reduced cross-region Iceberg source registration latency from 2+ minutes to sub-second by moving synchronous RPC provisioning to async workers and separating physical Iceberg locations from user-facing Snowflake views
- Improved self-service ingestion for 90+ analytics and engineering teams by simplifying source configuration, surfacing Snowflake table, view, and Iceberg metadata, and enabling table refreshes through an internal agent
- Reduced ingestion complexity and edge-case failures by introducing and backfilling explicit Snowflake object-type metadata, removing manual parsing and special-case logic while adding safeguards for table lifecycle operations

Data for Common Good June 2025 – August 2025
Software Engineer Intern *Chicago, IL*

- Developed a GraphQL-based ETL pipeline that consolidated data across 5+ pediatric cancer consortia
- Implemented a caching layer for AWS Lambda-queried datasets, reducing average response times by over 70%
- Reworked GraphQL data fetching to eliminate redundant requests, reducing API calls by 60%
- Refactored legacy code, improving readability and enabling easier integration across research workflows

Computer Science Instructional Laboratory March 2024 – May 2026
Software Developer *Chicago, IL*

- Developed SemaDoc, an in-house AI documentation tool used across 8 teams, cutting technical documentation turnaround from hours to minutes with LLM-powered workflows
- Architected SemaDoc's FastAPI backend, integrating async Trello APIs, Ollama-hosted LLMs, and MongoDB to support 500+ concurrent users with sub-100ms database operations
- Contributed to rewriting the CSIL website using Next.js and Node.js, serving 5,000+ UChicago students
- Built a secure alumni database with FastAPI, enabling streamlined data entry, search, and record management

Projects

CrustyDB | *Rust* Jan 2026

- Built the slotted-page layout and heap-file storage manager for CrustyDB, an academic relational database
- Implemented the aggregate, hash join, and nested-loop join operators in its volcano-style execution engine

Raft KV store | *Go, gRPC, Protocol Buffers* April 2026

- Implemented Raft from scratch, covering leader election, log replication, and commit-index advancement
- Layered a KV store over the consensus module, using gRPC and Protocol Buffers for the node-to-node interface

spotmv | **github** | *Python, Spotify Web API* June 2026

- Built a Python Spotify CLI for bulk playlist transfers, sorting, and renaming with dry-run safeguards for destructive actions

http-server | **github** | *Go* July 2025

- Built an HTTP/1.1 server from scratch with persistent connections, static file serving, custom routing, and concurrent request handling